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Scalable Extraction of Author-Stated Research Gaps: Towards Automated Idea Generation

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Abstract

Research gaps stated by authors could feed automated idea generation, but they are rare and scattered in papers. Asking a large language model (LLM) to judge every sentence, or a whole paper at once, is expensive, and long inputs are hard for smaller models. We propose a hierarchical three-stage funnel that splits the task: rules drop unlikely text, a zero-shot natural-language-inference classifier flags candidate gaps, and an LLM filters the candidates as source sentences. A single instruction sets the filter to favour precision, balance or recall. We evaluated the funnel on two human-labelled benchmarks, SAL (limitations in randomized-trial reports) and FWS (future work in natural language processing papers). Because each labels only one gap type in one field, we also built a corpus of 230 arXiv and biomedical papers in which two LLMs labelled limitations, future work and research gaps, with a third LLM pass resolving disagreements, and evaluated on it as well. In its high-precision setting, the LLM filter more than doubled precision on SAL, from 29.1% after the classifier to 66.2%, while keeping 84.9% recall, and on every dataset the three settings traded precision for recall as intended. The funnel used 3.7–13 times fewer LLM input tokens than judging every sentence.

Keywords: research gap extraction; limitations; future work; scientific text mining; large language models; natural language inference; zero-shot classification; LLM filtering

1. Introduction

Research papers often tell readers where a study falls short and what should be investigated next. In clinical research, self-acknowledged limitations help readers interpret findings in light of study design, data and applicability [1]. In other fields, future-work statements name extensions, open questions and further evaluations that authors consider worth pursuing [2,3]. Authors also point to unsolved problems in their field when they motivate a study. We call these statements *author-stated research gaps*: limitations of the authors' own work, future-work directions they propose, and open problems they identify. Collected across many papers, such statements are a candidate input for automated idea generation, because each one is a claim, made by a domain expert, about what is still missing.

Extracting these statements at scale is harder than it looks. Gap statements are sparse: in the corpora studied here they make up 2.6% of the sentences of full RCT reports, 5.7% of full arXiv and biomedical papers, and 14.6% of the sentences of conclusion sections. They

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are also scattered. Authors may put limitations in a dedicated section, inside a discussion, or next to a methodological choice, and in computer-science papers many gaps sit in the methods and experiments sections rather than at the end. A sentence that criticises prior work, restates a result or describes a mitigation can look like a gap without being one. Finally, the output must stay attributable: a system that generates plausible criticism is answering a different question from one that finds what the authors actually wrote [4,5].

Large language models (LLMs) can judge whether a sentence is a gap, but sending every sentence of a paper to an LLM is costly, and sending a whole paper leaves the model to find a few gap sentences in a very long input. Earlier extraction work suggests a cheaper division of labour. Clinical limitation recognition uses headings, lexical cues and supervised classifiers [1]; later work on randomized controlled trial (RCT) reports separates limitation detection from limitation typing [6]; and future-work recognition separates locating sentences from classifying them [2,3]. These studies motivate placing cheap structural and lexical evidence before more flexible semantic judgement.

We study a three-stage funnel built on this idea. Stage A discards text that is unlikely to contain gaps, Stage B flags candidate sentences with a classifier, and Stage C asks an LLM to filter the candidates. Every stage selects existing sentences and never rewrites them, so each output can be traced to its source. Two design choices are central to this study. First, Stage B is a zero-shot natural-language-inference (NLI) cross-encoder that needs no task-specific training labels, so the same funnel runs on biomedical and computer-science papers. Second, Stage C has three modes set by a single instruction: high precision, balanced and high recall. Because the extracted gaps are meant to feed idea generation, where a false gap wastes effort, we treat precision as the primary goal and use high precision as the main mode.

We evaluate the funnel on two benchmarks with human labels, randomized-trial reports with limitation labels (SAL) [6] and future-work sentences from natural language processing (NLP) papers (FWS) [3], and on a corpus of 230 arXiv and biomedical papers that we labelled for three gap types with an ensemble of LLMs. Results on the LLM-labelled corpus are reported as provisional until human validation, which is in progress, is complete.

Research questions.

- **RQ1 (Stage contribution).** How much does each stage change precision and recall on full-paper sentence streams?
- **RQ2 (Precision control).** Can a single instruction set the precision–recall trade-off of the LLM filter, and do the settings behave consistently across datasets?
- **RQ3 (Classifier and cost).** Does a training-free NLI classifier suffice for Stage B compared with supervised classifiers, and how much LLM input does the funnel save?

Contributions.

1. A source-preserving funnel for author-stated research gaps with a training-free Stage B and an LLM filter whose precision–recall trade-off is set by one instruction (Section 3).
2. An evaluation on two human-labelled benchmarks, including a check on SAL development papers on which no prompt was written (they were used only to set the classifier threshold), and on a 230-paper, three-type LLM-labelled corpus, with paper-level confidence intervals and paired comparisons (Section 4).
3. Evidence that the filter modes behave as intended on every dataset: high precision mode reaches 66.2% precision on SAL, 76.5% on FWS and 86.6% against the LLM labels, and the training-free classifier comes within 0.021 F_1 of a supervised PubMed-

BERT on SAL, a difference that is not statistically significant, although the supervised classifier is 8.4 points more precise.

4. Two findings about evaluation data: letter case alone reveals the label in the public FWS corpus, and SAL and our LLM labels disagree on most sentences the filter keeps that SAL marks negative (Sections 4.6 and 4.5).

The scope is extraction. Whether the extracted gaps are still open, and whether they lead to useful research ideas, are separate questions for future work.

Section 2 reviews related work, Section 3 describes data, funnel and evaluation, Section 4 reports results, Section 5 discusses them and their limitations, and Section 6 concludes.

2. Related Work

Our task connects the recognition of author-stated limitations and future work with rhetorical sentence classification and with LLM-based processing of scholarly text. These areas share input documents and some methods, but their output claims differ. Table 1 summarises the studies closest to ours.

2.1. Rhetorical annotation and scientific sentence classification

Argumentative zoning classifies sentences by their role in a scientific argument [7], and later schemes such as Core Scientific Concepts add categories related to gaps, failures and future work [8]. The Dr. Inventor corpus annotates limitations and future work in separate layers of 40 full articles [9]. Abstract-level resources such as PubMed 200k RCT, CSAbstruct, MAZEA and Emerald 110k broaden the available schemes [10–13], but an abstract role such as conclusion does not tell whether a sentence states a gap. Context helps classify short sentences: hierarchical and cross-domain models use surrounding sentences [14,15], LLMs have been applied to multi-label sentence classification [16], and context-enhanced models extract problem and method sentences [17]. Our filter also gives the LLM neighbouring sentences, but only to judge a candidate, not to label every sentence of a paper.

2.2. Author-stated limitations and future work

Kilicoglu et al. [1] recognise self-acknowledged limitations in clinical research with section rules and learned classifiers; their use of headings to narrow the search is a direct precedent for our Stage A. Lan et al. [6] annotate 200 RCT publications (SAL) and fine-tune PubMedBERT for limitation detection and typing, reporting a binary detection F_1 of 0.821 on a section-filtered candidate pool. That pool and its positive rate differ from the full-paper stream we score, so the numbers are not directly comparable. For future work, Zhu et al. [2] recognise future-research sentences in JASIST articles and cluster them, the ACL FWS-RC corpus provides future-work labels and a taxonomy [18], and Zhang et al. [3] report recognition and classification results on it (macro-F_1 0.907 for recognition). Suray et al. [19] trace whether later work engages with future-work statements, a downstream use of extraction.

2.3. LLM extraction, inferred gaps and generated critique

Several recent studies use LLMs on related tasks. LimGen collects author-written limitation sections from 4068 ACL papers to generate suggested limitations [20]. BAGELS combines parsing and LLM processing to extract and generate limitations across NLP, machine learning and biology [21], and FutureGen extracts future-work statements and then generates new ones with retrieval augmentation [22]. GAPMAP separates explicitly stated knowledge gaps from gaps inferred from context in biomedical text [23]. LimitGen

and Tree-of-Concerns ask models to find limitations that authors did not state [4,5]. In these works the LLM often generates text; in ours it only accepts or rejects a sentence the authors wrote. Sci-Sentence labels research gaps and limitations among seven rhetorical roles in literature reviews [24]; its research-gap class is close to our third type, but its sentences come from review sections rather than full papers.

2.4. Position of this work

Structural filtering, learned classifiers and LLM judgement each have precedents. What we add is a measured combination of the three on full-paper sentence streams, a training-free Stage B that transfers across domains, an explicit control of the LLM filter's precision–recall trade-off, and an evaluation that keeps label source, dataset and processing unit separate.

Table 1. Selected related work, ordered by year. "Unit" is the input each method processes.

Study	Domain	Statement type	Approach	Unit	Relation to this study
Dr. Inventor, 2016 [9]	Computer graphics	Limitations; future work	Human multi-layer annotation	Sentences, full articles	Full-text precedent with both statement types
Kilicoglu et al., 2018 [1]	Clinical research	Limitations	Rules; supervised classifiers	Sentences	Precedent for section-based localisation
Zhu et al., 2019 [2]	Information science	Future work	BERT classification; clustering	Sentences, full articles	Links recognition to thematic analysis
ACL FWS-RC, 2020 [18]	NLP	Future work	Annotation; taxonomy	Sentences	Source of FWS labels
Zhang et al., 2022 [3]	NLP	Future work	Lexical and neural recognition	Sentences	FWS recognition benchmark
LimGen, 2024 [20]	NLP	Limitations	LLM generation	Paper	Author limitation sections as references
Lan et al., 2024 [6]	RCT reports	Limitations	PubMedBERT detection and typing	Sentences	Source of SAL; supervised comparison
Lan et al., 2024 [16]	Several sciences	Rhetorical roles	LLM prompting and fine-tuning	Abstract sequence	Context-aware LLM labelling
Zhang & Zhang, 2024 [17]	NLP	Problem; method	Context-enhanced transformer	Sentence with context	Local context for sparse targets
Brack et al., 2024 [15]	Several sciences	Rhetorical roles	Cross-domain multi-task learning	Sentence sequence	Transfer across domains
Suray et al., 2024 [19]	Usable security	Future work	Statement and citation analysis	Statement	Downstream use of extracted directions
BAGELS, 2025 [21]	NLP; ML; biology	Limitations	Parsing; LLM extraction and generation	Paper	Closest extraction precedent
FutureGen, 2025 [22]	NLP; ML	Future work	LLM extraction; RAG generation	Paper	Extraction feeding generation
GAPMAP, 2025 [23]	Biomedicine	Knowledge gaps	LLM detection and inference	Paragraph to paper	Explicit versus inferred gaps
LimitGen, 2025 [4]	AI	Limitations	LLM and agent evaluation	Paper	Unstated limitations; different target
Sci-Sentence, 2025 [24]	Literature reviews	Research gap; limitation	Encoder and LLM classifiers	Sentence	Research-gap label close to ours
Tree-of-Concerns, 2026 [5]	ML; NLP; vision	Limitations	Multi-agent critique	Paper	Unstated critique; different target
This study	RCT reports; NLP; arXiv	Limitations; future work; research gaps	Rules, zero-shot NLI, LLM filter with three modes	Sentence with context	Measures stage contribution, precision control and cost

3. Materials and Methods

3.1. Task

The input is the sentence stream of a full paper. The output is the set of sentences that state an author-stated research gap of one of three types:

- **Limitation**: a weakness, caveat, bias, constraint or assumption the authors acknowledge about their own work;
- **Future work**: an explicit direction the authors propose for future research;

- **Research gap**: an unsolved problem, missing capability or open challenge the authors point to in the field or in prior work.

Each output is an unmodified source sentence. Scoring is binary at the sentence level: a predicted sentence is correct if its label is any gap type covered by the benchmark. SAL labels only limitations and FWS only future work, so on those benchmarks the funnel targets that one type.

3.2. Data

Table 2 summarises the evaluation data. All splits are by paper.

Table 2. Evaluation data. Positives are sentences labelled with a gap type the benchmark covers.

Dataset (split)	Labels	Gap types	Papers	Sentences	Positives	Use
SAL (test)	Human	Limitation	40	7968	205	Main evaluation
SAL (dev)	Human	Limitation	40	7729	180	Threshold; held-out check
FWS (test)	Human	Future work	209	1506	220	Evaluation
FWS (dev)	Human	Future work	418	3005	418	Threshold
Gold corpus (test)	LLM ensemble	All three	34	11,741	664	Evaluation
Gold corpus (validation)	LLM ensemble	All three	34	11,201	699	Design choices; thresholds

SAL.

SAL contains 200 RCT publications with sentence-level limitation labels, split into 120 training, 40 development and 40 test papers [6,25]. We use full papers, not a section-filtered pool. Section types were recovered by matching each sentence to the PubMed Central BioC version of its article, and sentences of fewer than three words were removed; none of the removed test sentences is a limitation. The training split is used only to train the supervised comparison classifiers.

FWS.

The FWS recognition corpus holds conclusion-section sentences from ACL papers labelled as future work or not [3,18,26]. Because it contains conclusion sections only, it has no full-paper structure. We use a fixed test sample of 209 papers (1506 sentences) and a development sample of the next 418 papers in the same random order, after removing sentences that appear in an earlier training set of ours. In this corpus every positive sentence is entirely lowercase and almost no negative is (Section 4.6), so we lowercase all FWS text before any processing.

Gold corpus.

We built a three-type corpus of 230 papers and 68,719 sentences: 190 arXiv computer-science and machine-learning papers, converted from PDF with PyMuPDF and split into sentences after removing reference lists, and the 40 SAL test papers with their original sentence boundaries. Two LLMs, Claude Opus 5.5 and Claude Sonnet 5.5, each labelled every sentence of every paper in one call per paper, following a written guideline with the three definitions above. Sentences marked by both with the same type form the high-confidence tier (2960 sentences). The other 1737 candidates marked by only one model, or with different types, were adjudicated by Claude Opus 5.5 with ± 3 sentences of context, which confirmed 1502. The final corpus has 4462 gap sentences: 2234 research gaps, 1543 limitations and 685 future-work statements. Agreement between the two labelling models on whether a sentence is a gap was $\kappa = 0.778$ [27]; this is agreement between models, not with humans. On the 40 SAL papers, the ensemble labelled all but nine of SAL's 205 human-labelled limitations as a gap. SAL labels were not shown to the labelling models.

Papers were split into 162 training, 34 validation and 34 test papers. The test split has 664 gaps (220 limitations, 97 future work, 347 research gaps); 28 test papers are arXiv (527 gaps) and 6 are SAL papers (137 gaps). Labelling cost about US$36.

3.3. The funnel

Figure 1 shows the three stages and the gold-test counts at each.

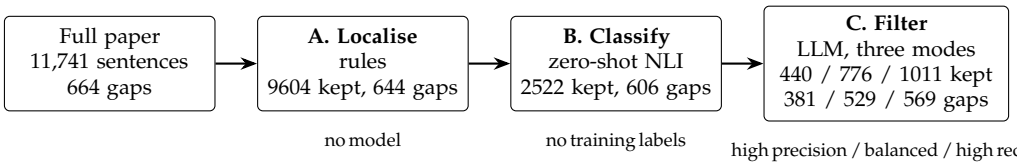


Figure 1. The funnel, with counts on the gold test split (34 papers, LLM labels). Each stage selects existing sentences; none rewrites text.

Stage A: localisation.

For SAL, Stage A keeps sentences in Discussion and Conclusion sections, which retains 203 of 205 test limitations while removing 76.3% of sentences. For the gold corpus, Stage A is a rule-based junk filter with no section gate: it drops titles, tables, figure captions, abbreviation lists, competing-interest, author-contribution, funding and supplementary sections where section types are known, bibliography-like strings, and implausible sentences (under 25 characters or six tokens, starting mid-word, dot leaders, or mostly non-letters). A section gate is not used for arXiv papers because their gaps are spread through the body (Section 5). FWS has no Stage A: it already consists of conclusion sections, and our junk filter, which rejects sentences that start in lower case, would reject almost all FWS text once it is lowercased.

Stage B: zero-shot NLI classifier.

Each sentence is scored by an NLI cross-encoder, ModernBERT-large fine-tuned on NLI data [28,29] (revision `ca476cb9`), used as a zero-shot classifier [30]. The sentence is the premise and a short hypothesis states the gap type: "This is a limitation.", "This is future work." or "This is a research gap." The score is the entailment probability divided by the sum of the entailment and contradiction probabilities. On the gold corpus a sentence is a candidate if its maximum score over the three hypotheses reaches a threshold τ, and its predicted type is the hypothesis with the highest score. On SAL only the limitation hypothesis is used, and on FWS only the future-work hypothesis. Inputs are truncated to 256 tokens.

Stage C: LLM filter.

Candidates are sent in reading order, 20 per request, to Qwen3.8-27B (FP8, revision `017b9c7a`), self-hosted with vLLM [31]. This model played no part in making any labels. Each item contains the predicted type, the sentence (up to 400 characters) and its context, the previous and next sentence of the same paper (up to 600 characters). The model returns a keep or drop verdict for each item as JSON constrained by a schema. Decoding uses temperature 0, seed 0 and no reasoning mode. The system prompt (Appendix A) gives the three definitions, a list of sentence kinds to drop (results, method descriptions, contribution claims, background, citations, captions and similar), and a final instruction that sets the mode:

- **High recall**: when genuinely uncertain, keep the sentence;
- **Balanced**: favour neither keeping nor dropping;

- **High precision**: keep a sentence only if it explicitly states the gap itself, drop general background, motivation, observed outcomes and passing remarks, and when uncertain, drop it.

For SAL and FWS the definitions are narrowed to the benchmark's single type, and the other two types are added to the drop list. The wording of each mode is otherwise identical across datasets.

3.4. How design choices were made

Because several choices were tuned, we record where each was made.

- **NLI hypotheses.** Hypothesis variants (long definitions, short type names, a single combined hypothesis, adding context, competing negative hypotheses) were compared on one half of the gold validation papers and confirmed on the other half. Short type names were best; context and negative hypotheses hurt.
- **Thresholds.** τ is the highest value whose overall recall, counting sentences lost in Stage A, is at least 90% on development data: gold validation ($\tau = 0.354$), SAL dev ($\tau = 0.7286$) and FWS dev ($\tau = 0.1922$). Values are rounded; at the rounded SAL value one development limitation falls below the threshold, so SAL dev recall before Stage C is 89.4% rather than 90.0%.
- **Stage C model and rubric.** The model, batch size, use of context and the high-recall rubric were chosen during development using 196 candidates from gold test papers; we report gold results with and without these sentences. The balanced and high-precision wording was written after reading false positives from a high-recall run on gold test papers, and was then checked on the gold validation split before any further test run. All three modes were run on every test set and are reported, so no mode was selected by its test score.
- **Held-out check.** The SAL dev papers were used only to set τ; no prompt was written or chosen on them. We run the full funnel on them as a check on unseen papers.

3.5. Comparison systems

Supervised classifiers on SAL.

We trained eight classifiers on the SAL training split: three fine-tuned encoders, PubMedBERT [32] (`BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`), SciBERT [33] and DistilBERT [34], each trained for three epochs (learning rate 3×10^{-5}, batch 32, class-weighted loss) with a threshold chosen on the SAL dev split; and five frozen sentence encoders [35,36] with logistic regression. PubMedBERT was the strongest and serves as the supervised Stage B comparison; it is also the model family used by Lan et al. [6].

Other Stage B options on the gold corpus.

A frozen bge-small encoder [36] with a logistic-regression head trained on the gold training split, and the same ModernBERT NLI model with a single long hypothesis covering all gap types (threshold 0.75 chosen on validation).

Whole-paper LLM prompting and published methods.

Comparisons with whole-paper and every-sentence LLM prompting on the same test papers, and reproductions of published methods, are in progress and are not reported in this version.

3.6. Evaluation

We report positive-class precision, recall and F_1 at the sentence level. The recall denominator is every positive sentence in the test papers, including those removed by Stage A.

Confidence intervals (CIs) are 95% percentile intervals from 2000 bootstrap resamples of papers [37], since sentences from one paper are not independent. Comparisons between two systems are paired: both are scored on the same resampled papers, and we report the interval of the difference. LLM input cost is estimated with the `cl100k_base` tokenizer [38] on the prompts and items exactly as sent; output tokens and Stage A and B computation are excluded. For comparison we count the input of one whole-paper call (the high-precision prompt followed by the numbered sentences) and of judging every sentence in batches of 20 in the funnel's format. We measured run-to-run stability by repeating the high-recall run on the gold test candidates.

3.7. Software and use of generative AI

Experiments used Python with PyTorch, Hugging Face Transformers, sentence-transformers, scikit-learn and vLLM; exact versions are pinned in the repository's lock file. Code is available at the repository listed in the Data Availability Statement. Generative AI is part of the method in two places: Claude Opus 5.5 and Claude Sonnet 5.5 produced the gold-corpus labels, and Qwen3.8-27B is the Stage C filter. Use of AI assistance in preparing the study and manuscript is declared in the Acknowledgments.

4. Results

4.1. Contribution of each stage (RQ1)

Table 3 and Figure 2 follow the positives through the funnel in high precision mode. Stage A removes most non-gap text in SAL at almost no recall cost, but little in the gold corpus, where it only drops junk. Stage B raises precision about threefold (2.7 to 3.6 times) while keeping about 90% of positives, as its threshold rule requires. Stage C then raises precision again, by 37 points on SAL, 28 on FWS and 63 against the gold labels, at a recall cost of 6, 12 and 34 points.

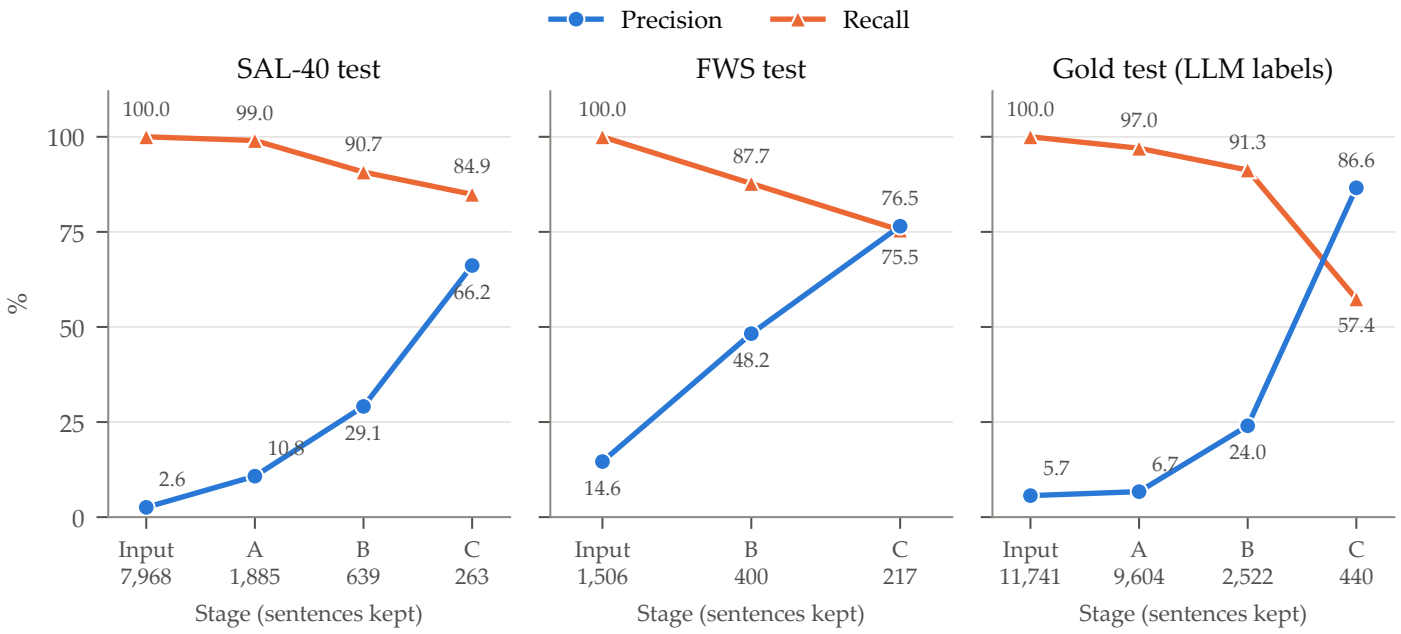


Figure 2. Precision and recall after each stage in high precision mode, with the number of sentences kept below each stage. Gold results use LLM labels; FWS has no Stage A.

Table 3. Sentences and positives kept after each stage, high precision mode. Precision and recall in %. Gold results use LLM labels.

Dataset	After stage	Sentences	Positives	Precision	Recall
SAL-40 test	Input	7968	205	2.6	100
	A. Discussion and Conclusion sections	1885	203	10.8	99.0
	B. NLI, $\tau = 0.7286$	639	186	29.1	90.7
	C. LLM filter	263	174	66.2	84.9
FWS test	Input (no Stage A)	1506	220	14.6	100
	B. NLI, $\tau = 0.1922$	400	193	48.2	87.7
	C. LLM filter	217	166	76.5	75.5
Gold test	Input	11,741	664	5.7	100
	A. Junk filter	9604	644	6.7	97.0
	B. NLI, $\tau = 0.354$	2522	606	24.0	91.3
	C. LLM filter	440	381	86.6	57.4

4.2. Precision control with three filter modes (RQ2)

Table 4 reports all three modes on every evaluation set, and Figure 3 plots them. On each set, precision rises and recall falls from high recall to balanced to high precision, as intended. High precision mode reaches 66.2% precision on SAL-40, 76.5% on FWS and 86.6% against the gold labels (83.4% on arXiv papers, 97.1% on the six biomedical papers).

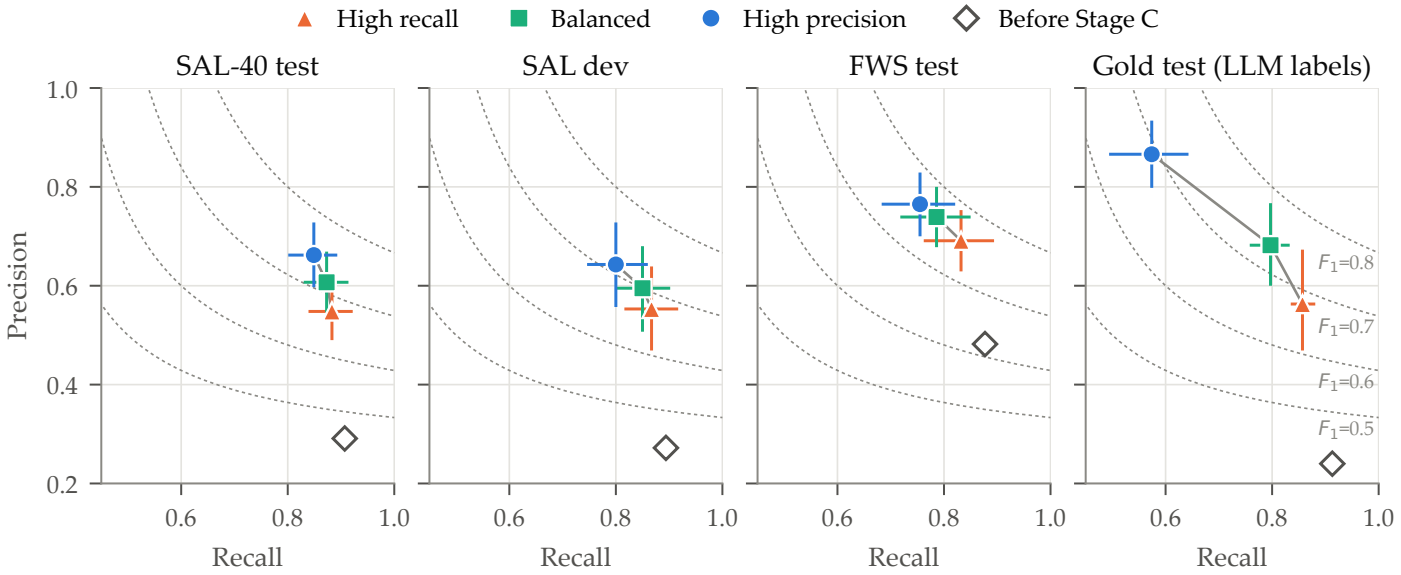


Figure 3. Precision against recall for the three filter modes, with 95% confidence intervals from resampling papers, and the Stage B output before filtering. Dotted curves mark equal F_1. Gold results use LLM labels.

Table 4. Results of the three filter modes. Precision and recall in %, with 95% confidence intervals from resampling papers. Gold results use LLM labels. Kept: sentences returned.

Evaluation set	Setting	Kept	Precision	Recall	F_1
SAL-40 test (human labels)	Before Stage C	639	29.1 (26.2–32.1)	90.7 (86.7–94.5)	0.441 (0.404–0.475)
	High precision	263	66.2 (59.7–72.8)	84.9 (80.1–89.3)	0.744 (0.695–0.789)
	Balanced	295	60.7 (54.6–66.9)	87.3 (83.0–91.4)	0.716 (0.670–0.760)
	High recall	330	54.8 (49.0–61.3)	88.3 (83.9–92.2)	0.677 (0.627–0.725)
SAL dev (human labels)	Before Stage C	591	27.2 (23.3–31.5)	89.4 (85.1–93.5)	0.418 (0.369–0.468)
	High precision	224	64.3 (55.7–72.8)	80.0 (74.6–86.0)	0.713 (0.647–0.774)
	Balanced	257	59.5 (50.7–68.0)	85.0 (79.9–90.2)	0.700 (0.628–0.764)
	High recall	282	55.3 (46.9–63.9)	86.7 (81.6–91.7)	0.675 (0.602–0.742)
FWS test (human labels)	Before Stage C	400	48.2 (42.2–54.5)	87.7 (81.1–93.4)	0.623 (0.565–0.677)
	High precision	217	76.5 (70.0–82.9)	75.5 (68.3–82.1)	0.760 (0.705–0.807)
	Balanced	234	73.9 (67.8–80.0)	78.6 (71.8–85.0)	0.762 (0.709–0.808)
	High recall	265	69.1 (62.9–75.3)	83.2 (76.2–89.4)	0.755 (0.702–0.804)
Gold test, all (LLM labels)	High precision	440	86.6 (79.8–93.4)	57.4 (49.4–64.3)	0.690 (0.619–0.748)
	Balanced	776	68.2 (60.0–76.7)	79.7 (75.8–83.3)	0.735 (0.678–0.789)
	High recall	1,011	56.3 (46.9–67.3)	85.7 (83.5–88.1)	0.679 (0.604–0.756)
Gold test, arXiv (LLM labels)	High precision	337	83.4 (76.0–91.3)	53.3 (44.4–61.1)	0.650 (0.573–0.713)
	Balanced	643	64.2 (56.0–73.6)	78.4 (73.8–82.6)	0.706 (0.647–0.765)
	High recall	862	51.7 (42.8–62.7)	84.6 (82.1–87.2)	0.642 (0.567–0.719)
Gold test, biomedical (LLM labels)	High precision	103	97.1 (94.5–99.0)	73.0 (63.7–81.0)	0.833 (0.761–0.885)
	Balanced	133	87.2 (82.4–92.2)	84.7 (80.5–87.7)	0.859 (0.833–0.889)
	High recall	149	82.6 (80.2–86.1)	89.8 (85.7–94.4)	0.860 (0.839–0.894)

The ranking of the modes by F_1 depends on the dataset. On SAL-40 high precision has the highest F_1 and on FWS the three are within 0.007. On the gold corpus balanced is highest: its F_1 exceeds high recall by 0.055 (paired 95% CI 0.025 to 0.080) and high precision by 0.045 (0.006 to 0.095), while high precision and high recall do not differ ($+0.011$, -0.054 to $+0.060$). The SAL dev papers, which no prompt was written on, show the same pattern as SAL-40 at slightly lower values: high precision gives 64.3% precision, 80.0% recall and F_1 0.713.

The modes behaved the same way on the gold validation split, where they were checked before the balanced and high-precision test runs: in high precision mode the filter kept 60.3% of the gaps it received and removed 97.7% of non-gaps, against 94.1% and 82.9% in high recall mode, giving precision of 90.4% and 66.8% respectively.

Robustness.

On the gold test split, excluding the 196 candidates used during development changes high precision F_1 from 0.690 to 0.675, balanced from 0.735 to 0.723 and high recall from 0.679 to 0.664. Repeating the high-recall run on the 2522 gold test candidates changed 32 verdicts (1.3%), five of them on gaps, and F_1 moved from 0.678 to 0.679.

Figure 4 and Table 5 show the cost of precision by gap type on the gold corpus. Future-work recall is stable across modes (82.5–86.6%), while research-gap recall falls from 89.3% to 51.3% and limitation recall from 79.5% to 55.9%. Research gaps make up half of the gold labels and are the easiest type for the earlier stages, so scores that pool all types depend on them. Scored without research gaps, high precision mode is the best of the three (F_1 0.701 against 0.595 and 0.509).

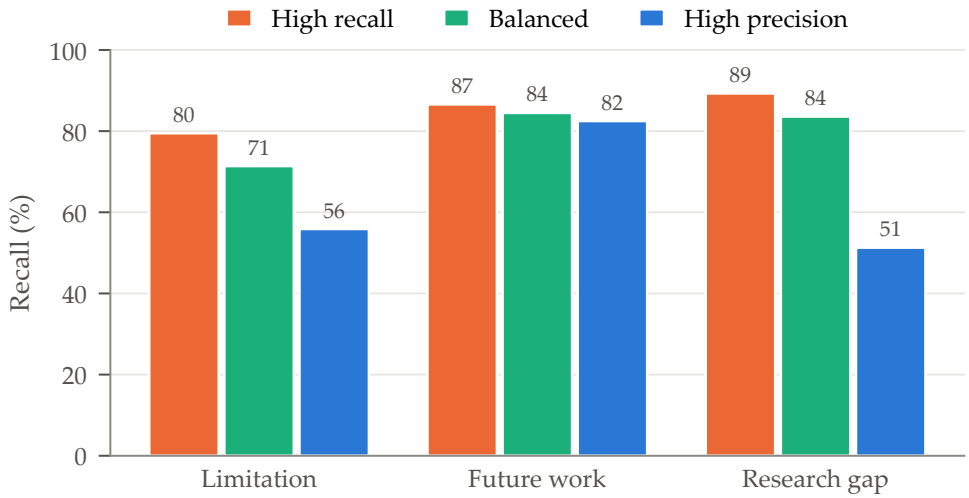


Figure 4. Recall by gap type on the gold test split (LLM labels) in each filter mode, in %.

Table 5. Gold test split (LLM labels) by gap type: recall per type, and scores when research-gap sentences are excluded from scoring. In % except F_1.

	Recall by type			Without research gaps		
Mode	**Limitation (220)**	**Future work (97)**	**Research gap (347)**	**Precision**	**Recall**	F_1
High precision	55.9	82.5	51.3	77.5	64.0	0.701
Balanced	71.4	84.5	83.6	49.2	75.4	0.595
High recall	79.5	86.6	89.3	36.9	81.7	0.509

4.3. Training-free versus supervised classification (RQ3)

On SAL-40, we ran the same three modes with the supervised PubMedBERT classifier as Stage B. Before Stage C, PubMedBERT gives 46.0% precision and 87.8% recall (F_1 0.604) on 391 candidates, the best of eight SAL-trained classifiers (SciBERT 0.553, DistilBERT 0.523, frozen encoders 0.371–0.457). After Stage C in high precision mode it reaches 74.5% precision (95% CI 68.4–80.7), 78.5% recall and F_1 0.765. Table 6 compares it with the NLI classifier. With PubMedBERT the funnel is 6.9 to 9.0 points more precise in every mode, but in high precision mode the F_1 difference is not significant (−0.021, CI −0.053 to +0.014). The NLI classifier used no SAL labels at all; its only SAL-specific setting is the threshold.

Table 6. SAL-40 test. Left: NLI minus PubMedBERT as Stage B, same mode, paired over resampled papers (precision in points; 95% CI in brackets). Right: precision of the NLI funnel's output when scored against the LLM ensemble's labels instead of SAL's human labels, in %.

	NLI minus PubMedBERT		Precision against LLM labels	
Mode	**Precision**	F_1	**Limitation**	**Any gap type**
High precision	−8.4 (−11.8, −4.5)	−0.021 (−0.053, +0.014)	93.2	95.1
Balanced	−6.9 (−11.5, −2.3)	−0.035 (−0.068, 0.000)	88.8	91.2
High recall	−9.0 (−12.7, −5.1)	−0.059 (−0.089, −0.027)	81.8	85.2

On the gold corpus, the NLI classifier also compares well with other training-free and lightly trained options. On the validation split, the three short hypotheses reached about 79% recall at 39.2% precision, where a bge-small head trained on the gold training split reached 28.2% precision at matched recall. On the test split, the NLI funnel in high precision mode exceeds the same ModernBERT model used alone with a single long hypothesis (F_1

0.614) by 0.076 (CI 0.020 to 0.132), and does not differ significantly from a funnel with the bge-small head and the high-recall filter (F_1 0.665; difference +0.025, CI −0.037 to +0.087).

4.4. Cost

Table 7 and Figure 5 estimate LLM input tokens in high precision mode; the other modes differ by under 2% because only the final instruction changes. Compared with judging every sentence, the funnel sends 3.7 to 13.1 times fewer tokens. Compared with a single whole-paper call, the saving is smaller: 3.1 times on SAL-40, 2.0 times on FWS and 1.25 times on the gold corpus. On arXiv papers, Stage B passes about one sentence in five, and each candidate carries its context, so the funnel sends almost as much text as the whole paper. Whether this is worthwhile depends on whether whole-paper prompting finds the same gaps, which the comparison in progress will show.

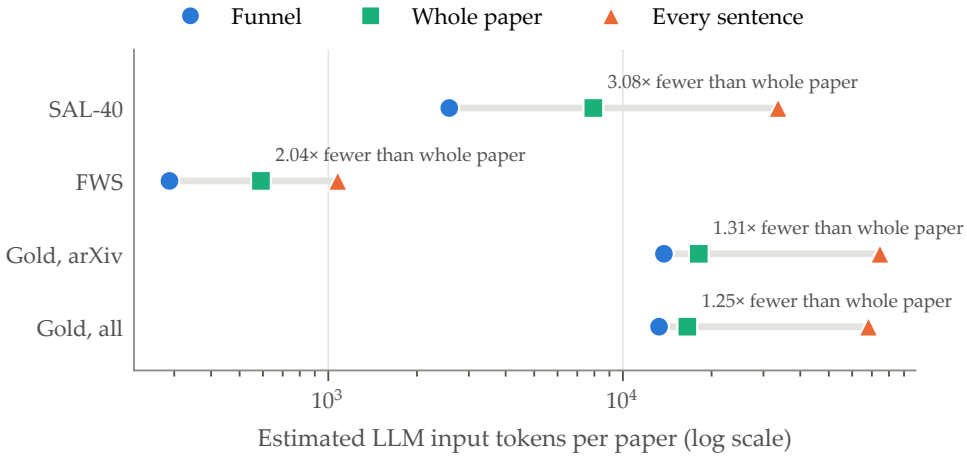


Figure 5. Estimated LLM input tokens per paper (`cl100k_base`, log scale) for the funnel in high precision mode, one whole-paper call, and judging every sentence in batches.

Table 7. Estimated LLM input tokens (`cl100k_base`), high precision mode, all test papers. Savings are the alternative's tokens divided by the funnel's tokens.

Dataset, Stage B	Candidates	Calls	Funnel	Whole paper	Every sentence	Saving vs whole paper	Saving vs every sentence
SAL-40, NLI	639	32	102,978	317,475	1,345,409	3.08	13.07
SAL-40, PubMedBERT	391	20	62,835	317,475	1,345,409	5.05	21.41
FWS, NLI	400	20	60,410	123,445	225,015	2.04	3.72
Gold test, NLI	2,522	127	450,955	562,440	2,318,416	1.25	5.14
Gold test (arXiv), NLI	2,137	107	385,897	506,753	2,087,907	1.31	5.41

4.5. Label source

The right side of Table 6 scores the same SAL-40 output against two label sets. Of the 263 sentences returned in high precision mode, SAL's annotators marked 174 as limitations, while the LLM ensemble marked 245 as limitations and 250 as some gap type. Seventy-seven sentences are gaps by the LLM labels but not by SAL; only 12 are negative under both. Precision is therefore 66.2% against SAL and 93.2–95.1% against the LLM labels. Which label set is right on these 77 sentences decides the true precision, and our human study (Section 5) is designed to settle it.

4.6. Letter case reveals the label in FWS

In the FWS recognition corpus as we obtained it (64,896 sentences from 9013 papers), every one of the 9009 positive sentences is entirely lowercase, and almost none of the negatives is. The rule "a sentence is positive if it is all lowercase" therefore scores 99.8% precision and 100% recall. Our copy has the same size in bytes as the published file [26], so the artefact appears to be in the released corpus, although we did not compare the two files directly. Any case-sensitive model, and any LLM given the raw text, can exploit it. We lowercased all FWS text for the results above, and we do not use our earlier FWS result on raw text.

5. Discussion

5.1. Answers to the research questions

RQ1: stage contribution.

The three stages do different work. Stage A is cheap and, when sections are meaningful as in RCT reports, removes three quarters of the text at almost no recall cost. Stage B sets the recall ceiling, about 88–91% in our runs, because the threshold rule fixes it on development data. Stage C does most of the precision work, raising precision by 28 to 63 points. Every reported gap is still a sentence the authors wrote.

RQ2: precision control.

A single change to the final instruction moves the filter along the precision–recall curve in the expected order on every evaluation set (SAL-40, SAL dev, FWS, and the gold test split overall and in its arXiv and biomedical parts), and the same order held on the gold validation split before the balanced and high-precision modes were run on test papers. This makes the funnel adjustable to its use: high precision suits idea generation, where a false gap costs a downstream model or a person time, while high recall suits a reviewer who wants to see every candidate. On human labels, high precision mode does not lose F_1: it is the best mode on SAL-40 and level on FWS. On the LLM-labelled corpus it loses recall, mostly on research gaps, which suggests that the high-precision wording, which asks for an explicit statement of the gap, is strictest with the statements authors use to motivate their own work.

RQ3: classifier and cost.

A zero-shot NLI classifier with short hypotheses naming the gap type is a workable Stage B. It needs no training labels, transfers from biomedical to computer-science text, and on SAL comes within 0.021 F_1 of a supervised PubMedBERT that was trained on SAL's own labels. The supervised classifier is more precise, by 8.4 points in high precision mode, so a project with in-domain labels should still use them. The cost saving over judging every sentence is 3.7 to 13 times. The saving over a single whole-paper call is modest and on arXiv nearly vanishes. A whole-paper prompt could also carry the mode instruction and return sentence numbers, so on arXiv the funnel is worthwhile only if it finds gaps more accurately than whole-paper prompting; the comparison in progress will show whether it does.

5.2. How good are the extracted gaps?

The answer depends on the labels. On SAL-40, high precision mode reaches 66.2% precision against SAL's labels and 93.2% against the LLM labels (Section 4.5). Either label set may be wrong on the 77 sentences where they disagree: SAL's annotators may have left some limitation statements unlabelled, and the ensemble's labels are inclusive by construction, since adjudication added sentences marked by only one model. True precision

is therefore likely to lie between the two readings. A development observation, not part of the results above, suggests that some gold-corpus false positives are also label errors: in an earlier high-recall run, five of the 34 test papers held 59% of the false positives, and each of them had only about 18–24 gold gaps despite having 220 to 824 sentences, which may indicate under-labelling of long papers. We have designed a blind human study with two raters and an independent tie-breaker on a frozen sample of 135 sentences, drawn from the high-precision output on arXiv and SAL-40 and from decoys, to measure true precision against people. Its results will be added in a later version.

5.3. Localisation does not transfer from clinical to computer-science papers

Section structure helps on RCT reports, where limitations cluster in Discussion and Conclusion sections. On arXiv papers it did not help in our development work. An earlier localiser that kept only sentences in sections where gaps are usually stated, such as the discussion and conclusion, kept 55.6% of the arXiv gold gaps with sections inferred from the extracted text and 55.2% with sections recovered from the LaTeX sources of 175 papers; 82% of the gaps it missed were in methods, experiments or results sections. These are development observations from a single run that we have not recomputed for this version. We therefore use only a junk filter as Stage A for arXiv papers, which keeps 97% of gaps, and leave narrowing to Stage B.

5.4. Evaluation data needs checking

The FWS case artefact (Section 4.6) shows that a published benchmark can carry a trivial signal that a modern model will find. Results on FWS from case-sensitive models or LLMs given raw text, including our own earlier ones, should be treated with caution until the text is normalised. More generally, our results differ by label source more than by method (Section 4.5). Reporting which labels a number rests on is as important as reporting the number.

5.5. Limitations

- **LLM labels.** All gold-corpus results rest on labels from an ensemble of Claude models; human validation is designed and its sample frozen, but not yet done. The inter-model agreement of $\kappa = 0.778$ does not measure accuracy.
- **Development on test papers.** The Stage C model, batch size, context use and high-recall rubric were chosen using 196 gold test candidates, and the balanced and high-precision wording was written after reading gold test false positives. We report results without the 196 candidates, ran all modes everywhere, and checked the modes on validation first, but the gold test split is not untouched. The SAL dev papers are the cleanest check, and they agree with SAL-40.
- **Scope of the benchmarks.** SAL covers RCT reports only, FWS covers NLP conclusion sections only, and each labels one gap type. The six biomedical gold papers are also SAL papers.
- **Research gaps.** Half of the gold labels are research gaps, which score higher than the other types in the earlier stages; we therefore report per-type recall and scores without them.
- **Comparisons in progress.** Whole-paper LLM prompting on the same papers and reproductions of published methods are not yet included.
- **Cost.** Token counts are estimates with a vendor-neutral tokenizer and cover LLM input only.
- **One LLM.** Stage C uses one model; results with other models may differ. Repeating one run changed 1.3% of verdicts.

6. Conclusions

We presented a three-stage funnel that extracts author-stated research gaps as unmodified source sentences: rules discard unlikely text, a zero-shot NLI classifier flags candidates without training labels, and an LLM filter with three instruction-set modes removes false positives. On human-labelled limitation and future-work benchmarks, high precision mode reached 66.2% and 76.5% precision at 84.9% and 75.5% recall, results confirmed on held-out SAL papers; against LLM labels on a 230-paper, three-type corpus it reached 86.6% precision at 57.4% recall. The modes moved precision and recall in the intended direction on every dataset, and the training-free classifier came close to a supervised one on SAL. The funnel's cost advantage is clear against judging every sentence and modest against whole-paper prompting. Next steps are the human validation of the corpus, comparisons with whole-paper prompting and published methods, and the use of the extracted gaps for automated research-idea generation.

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Abbreviations

The following abbreviations are used in this manuscript:

CI	Confidence interval
FWS	Future work sentence (corpus)
LLM	Large language model
NLI	Natural language inference
NLP	Natural language processing
RCT	Randomized controlled trial
SAL	Self-acknowledged limitations (corpus)

Appendix A. Stage C prompt

The system prompt used on the gold corpus is reproduced below. The modes differ only in the paragraph marked [MODE]. For SAL and FWS, the decision rule keeps only item (a) or (b) respectively, the other two items are added to the list of sentences to drop, and the words "gap" and "research gap" are replaced by "limitation" or "future-work direction". All prompts are in the code repository (`scripts/bench/stage_c_rubric_val.py`, `stage_c_sal40_modes.py`, `nli_sal_fws.py`).

> You are a precision filter for a research-gap extraction pipeline. Judge each candidate sentence: does it genuinely state a research gap?
>
> Each item has: “i” (integer id), “type” (the predicted gap type), “sentence” (the candidate to judge), and “context” (surrounding text). Judge the “sentence”, using “context” only to disambiguate.
>
> Decision — answer YES (keep) if the sentence states ANY of: (a) a limitation / weakness / caveat / bias / constraint / assumption of the authors’ OWN work (e.g., small sample, no blinding, single setting, short follow-up, confounding, limited generalizability, unvalidated measure, “we did not / could not . . . ”, “a limitation is . . . ”); (b) an explicit FUTURE-WORK direction the authors propose (“in future work we will . . . ”, “future research should . . . ”, “it would be interesting to . . . ”); (c) an unsolved problem / missing capability / open research gap in the field or prior work (“it remains unknown . . . ”, “no existing method . . . ”, “current approaches struggle with . . . ”).
>
> Answer NO (drop) if the sentence is instead: a result or finding, a method/experimental-setup description, a positive or contribution claim (“we propose/show/achieve . . . ”), background or related-work description, a citation or reference fragment, acknowledgements / funding / author contributions / competing interests, a figure or table caption, a dataset/metric definition, or a generic non-gap statement.
>
> [MODE]
>
> Return JSON of the form {“results”:[{“id”:<i>, “keep”:<bool>}, . . .]} with EXACTLY one entry per input item, where id is the item’s “i” and keep=true means YES.

The [MODE] paragraph:

- **High recall:** “Be KEEP-BIASED: when you are genuinely uncertain whether a sentence is a real gap, answer YES. Only answer NO when it clearly is NOT a gap by the rules above.”
- **Balanced:** “Judge each sentence on the rules above, without favouring YES or NO.”
- **High precision:** “Be STRICT. Answer YES only when the sentence itself explicitly states the limitation, the proposed direction, or the unsolved problem. Also answer NO for: a general property or difficulty of a method or domain given as background or motivation; a statement of why the authors’ approach is needed; an observed experimental outcome; and a passing remark that names no specific gap. When genuinely uncertain, answer NO.”

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